

Encoding verse inscriptions in MAPPOLA

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Sunoikisis Digital Classics – 30 June 2026

Digital Editions of Text-Bearing Objects (Seals, Inscriptions etc.)

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What is an inscription?

A text traced upon some hard substance for the sake of durability



Epigraphers operate on multiple levels

- **Archaeology**

- Type of monument
- Function
- Material
- Decoration
- Provenance

In the EpiDoc schema



Metadata in TEI Header

- **Text**

- Transcription
- Apparatus Criticus
- Other editions



Text

- **History**

- Prosopography
- Historical events mentioned in the text



Semantic metadata

```
<TEI>
<teiHeader>...</teiHeader>
<facsimile>...</facsimile>
    <text>...</text>
</TEI>
```

EpiDoc Template:

<https://raw.githubusercontent.com/EpiDoc/Source/main/guidelines/xml/ex-epidoc-template.xml>

E
<EpiDoc>

XML text markup for ancient documents

27 followers Crossing the Tiber, and Beyond <http://epidoc.sf.net>

Overview Repositories 13 Projects Packages People 1

Source Public

Main source code for EpiDoc (schema, guidelines and stylesheets)

● XSLT ☆ 10 📄 6

EFES Public

Forked from kcl-ddh/kiin

EFES (EpiDoc Front End Services) is a custom and readily customizable platform for publication and search/indexing of EpiDoc files, based on the Kiln platform

JavaScript ☆ 35 🍷 49

EpiDagito Public

EpiDoc / Recogito integration

☆ 3


[Stylesheets](#)
[Public](#)

The Reference Stylesheets developed and released by EpiDoc for use with XML documents following the EpiDoc schema.

● XSLT ☆ 19 📄 17

Tutorials

Online Syllabus for EpiDoc training

☆ 21 ♀ 3

ExampleScripts Public

Freestanding or exemplary scripts that might be useful for EpiDoc users. All content offered without guarantee.

● XSLT ☆ 2

EpiDoc Guidelines 9.8

EpiDoc Guidelines: Ancient documents in TEI XML

(This page in Bulgarian | Spanish | Italian | Portuguese)

EpDoc is an international, collaborative effort that provides guidelines and tools for encoding scholarly and educational editions of ancient documents. These pages are the EpDoc Guidelines produced by the collaborative. In addition, the [EpDoc Website](#) provides access to other tools and collaboration environments supported by the collaborative.

EpIDoc specifies a subset of the Text Encoding Initiative's standard for the representation of texts in digital form using the Extensible Markup Language (XML), a technical standard promulgated by the World-Wide Web Consortium. It addresses not only the transcription and editorial preparation of the texts themselves, but also the history, materiality and metadata of the objects on which the texts appear.

These Guidelines contain descriptions of the lexical, descriptive, and other features expressed with EpiDoc, as well as the TEI elements and attributes used to encode them. The pages below provide a number of entry points into the full documentation, via a series of thematic indexes into the content of the Guidelines, rather than a single "table of contents" reflecting a (nonexistent) ordered, hierarchical document structure.

Browse Guidelines

- [How to use the EpiDoc Guidelines](#) (bq) (de) (es) (fr) (it) (nl) (pt-br)
- [Overview of EpiDoc structure](#)
- [EpiDoc homepage](#)

[Tables of contents by transcription style:](#)

- Epigenetic information and descriptive data following PETRAE
- Developmental transcription following [POTTER](#)
- Phenological calendar data following [AP3S](#)
- Epigenetic transcription following [Cammer-Pancore 1999](#)
- Epigenetic transcription following [Pancore 1999](#)
- Psychological data following [HCS](#)
- Leaf transcription following [Lysiology](#) ([Hemiphytic](#) etc.)
- Leaf Lateralisation following [DGCs](#) ([Carnegie](#) coding)
- Text transcription following [Nissenbaum](#) using [Nissenbaum Conventions](#)
- Text transcription following [CEPAC](#) ([Archives](#) stamps)

Thematic lists of guidelines pages:

- [Descriptive and historical data](#)
- [Text transcripts](#)
- [Images and figures](#)
- [Vocabularies and index no terms](#)

Appendices

- [Gentle Introduction To Markup for Encipherers \(bg\) \(de\) \(es\) \(fr\) \(it\) \(pl\)](#)
- [Bibliography](#)
- [License \(bg\) \(es\) \(fr\) \(it\) \(pl\) \(pt\)](#)

Externally maintained guidelines for specialized communities

- [EniDoc: Leiden Quick Reference](#) (used in training workshops)
- [Shorter Leiden and EniDoc Quick Reference](#) (shorter version)
- [Leiden Structure Quick Reference](#) (used in training workshops)
- [Paypal: eniDoc/Leiden - help and examples](#)
- [Digital Manuscript Historiography: Manuscript encoding guidelines](#) (Miriam Panti et al., Leipzig)
- [Local Guidelines from Leiden projects \(InsAph, InsLip, IOSPE etc.\)](#)
- [TEI Encoding in Egyptology and Cuneology](#) (Daniel Worman, HU Berlin)

What is a verse inscription?

- An inscription containing verses or consisting entirely of a poem.
- An inscription containing metrical clauses or other poetic phrases, even if not full verses.
- More broadly: any form of written verbal art.

D(is) M(anibus)
M(arci) Romani Iovini,
rhetoris eloquii Latini.
Conditus hac Romanus
est tellure Iovinus,
docta loqui doctus
quique loqui docuit.
Manibus infernis,
si vita est gloria vitae,
vivit et hic nobis
ut Cato vel Cicero.

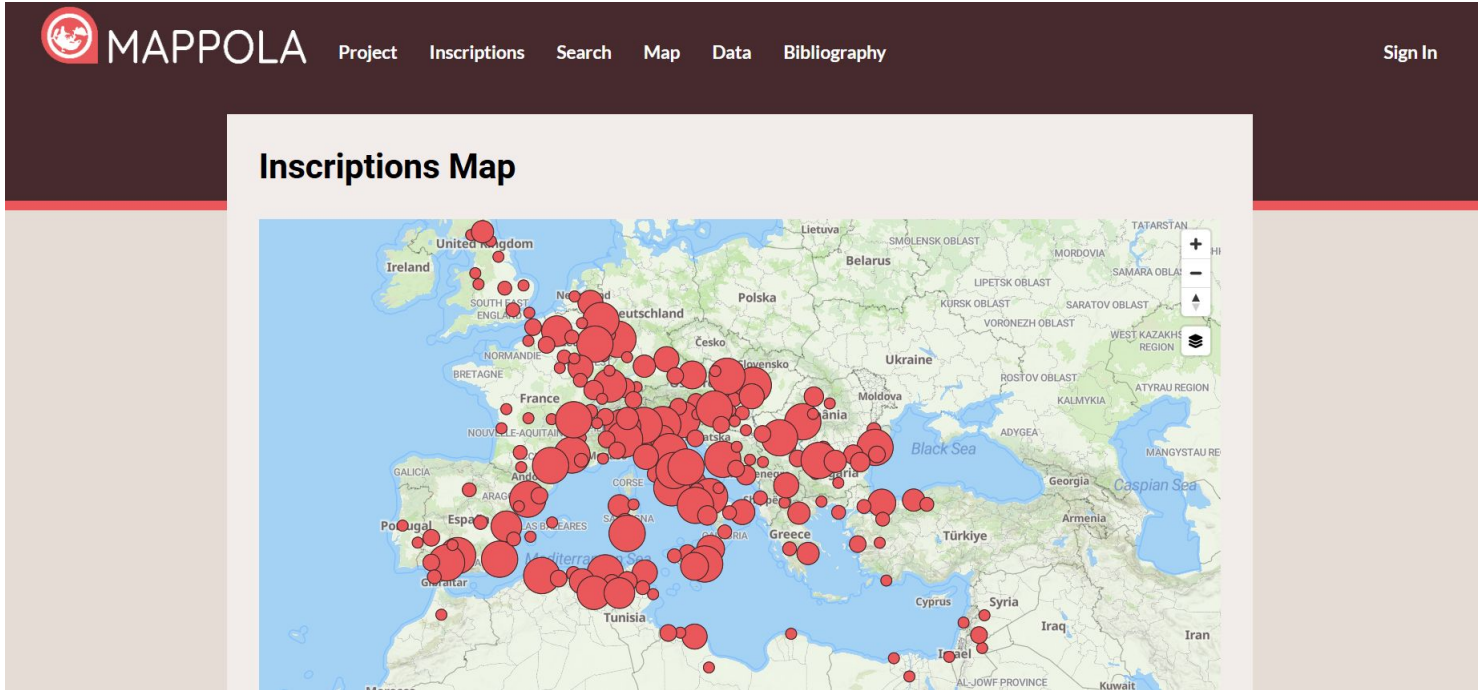
M(arcus) Iunius Severus et
Romania Marcia,
heredes, bene merenti
Fecerunt.



Roma, S. Paolo fuori le mura. Source: Wikimedia Commons

MAPping Out the POetic LAndscape(s) of the Roman Empire

MAPPOLA Database (<https://db.mappola.eu/>)



MAPPOLA in brief

- **Main goal**
 - To collect Greek and Latin **verse and rhythmical inscriptions** from the Roman Empire and make them searchable (currently circa 1000 complete entries out of more 4000)
 - To enable combined searches of metrical patterns historical features
- **Research questions**
 - Regional and ethnic diversity
 - **Uses of different social strata**
 - Microhistories and multilingual contexts
 - **Interdependency of poetic forms and socio-cultural diversity**
 - Transformations and reinvention of the production and consume of poetry

How the database works

- Hybrid architecture combining a relational and a non-relational database
 - Interactive map with two layers: present-day geography based on **OpenStreetMap** and the **Digital Atlas of the Roman Empire**
 - Relational database implemented in **MySQL**
 - EpiDoc markup is used **exclusively** for the transcription of the epigraphic text
 - Integration of the **Patrimonium EpiDoc converter** to generate EpiDoc compliant XML (<https://patrimonium.huma-num.fr/atlas/epidoc-converter/>)
 - Output available in **XML** format

```
from . import db
from sqlalchemy import event
from ..linkage import LINKERS
from ..linkage.epidoc import *
import json
import re, sys, flask
from flask import Markup
from lxml import html

class Inscriptions(db.Model):
    id = db.Column(db.Integer(), primary_key=True)
    title = db.Column(db.String(180))
    trismegistos_nr = db.Column(db.String(40))

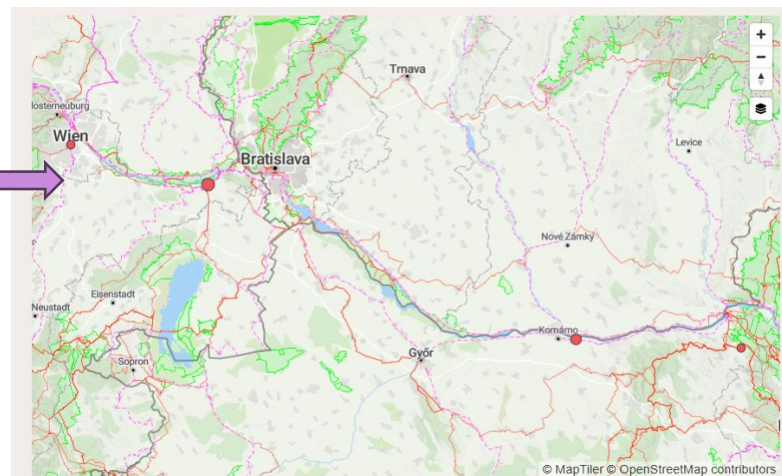
    place_id = db.Column(db.Integer, db.ForeignKey('places.id'))
    find_comment = db.Column(db.Text)
    coordinates_long = db.Column(db.Float())
    coordinates_lat = db.Column(db.Float())
    current_location_id = db.Column(
        db.Integer, db.ForeignKey('current_locations.id'))
    current_location_inventory = db.Column(db.String(80))
    current_location_details = db.Column(db.Text)

    object_type_id = db.Column(db.Integer, db.ForeignKey('object_types.id'))
    object_material_id = db.Column(
        db.Integer, db.ForeignKey('object_materials.id'))
    object_preservation_state_id = db.Column(
        db.Integer, db.ForeignKey('object_preservation_states.id'))
    object_dim_height = db.Column(db.Float) # Y
    object_dim_width = db.Column(db.Float) # X
    object_dim_depth = db.Column(db.Float) # Z
    object_execution_technique_id = db.Column(
```


What the MAPPOLA database provides

- **Interactive map and customizable filters**
- Verse visualisation
- Translations in multiple modern languages
- Layout description
- Prosopographical data
- Integrated search across textual, metrical, archaeological and historical features

Search Clear	
Mappola No.	MPL
province	Pannonia superior
country	
verse type	Dactylic hexameter
date	from to
bibliography	
inscription text	 AND
full text search	



You got 7 result(s).

- [Funerary stele for C. Valerius](#)
- [Stele for Vibia Clitheris](#)
- [Funerary inscription for Lucius Cominius Firmus](#)
- [Literary quote on a brick](#)
- [Funerary inscription for Valerius Ursinus and others](#)
- [Christian inscription on a tile](#)
- [Epitaph from Vindobona](#)

What the MAPPOLA database provides

- Interactive map and customizable filters
- **Verse visualisation**
- Translations in multiple modern languages
- Layout description
- Prosopographical data
- Integrated search across textual, metrical, archaeological and historical features

Diplomatic	Interpretative	Verse Visualisation
D(is) M(anibus) M(arci) Romani Iovini rhetoris eloquii Latini. paragraphos Conditus hac Romanus est tellure Iovinus, docta loqui doctus quique loqui docuit. Manibus infernis, si vita est gloria vitae, vivit et hic nobis ut Cato vel Cicero. paragraphos M(arcus) Iunius Severus et Romania Marcia, heredes, bene merenti fecerunt.		

Diplomatic	Interpretative	Verse Visualisation
	D(is) M(anibus) M(arci) Romani Iovini rhetoris eloquii Latini. paragraphos	
5	Conditus hac Romanus est tellure Iovinus, docta loqui doctus quique loqui docuit. Manibus infernis, si vita est gloria vitae, vivit et hic nobis ut Cato vel Cicero. paragraphos	
10		
15	M(arcus) Iunius Severus et Romania Marcia, heredes, bene merenti fecerunt.	

What the MAPPOLA database provides

- Interactive map and customizable filters
- Verse visualisation
- **Translations in multiple modern languages**
- Layout description
- Prosopographical data
- Integrated search across textual, metrical, archaeological and historical features

Inscription & Translations

Diplomatic

Interpretative

Verse Visualisation

Caelestis mulsa ((interpunct)) propria placabitur ara
hac divis aliis tura merumq(ue) damus.

Apparatus

Text follows Cenati, Murzea, González Berdús 2023.

Main Translation

Translation	Caelestis will be appeased with her own mead. On this altar we give to the other gods incense and wine.
Citation	Cenati, Murzea, González Berdús 2023.

Other Translations

- Die Himmlische Göttin wird auf ihrem eigenen Altar mit Weinmet besänftigt werden. Auf diesem (Altar) bringen wir (den) anderen Göttern Weihrauchopfer und (ungemischten) Wein dar. (German, Piso 1993)
- Caelestis will be appeased on the altar with her characteristic mead and to the other gods we give incense and wine. (English, González Berdús 2023)

Altar for Caelestis from Sarmizegetusa, Dacia (MPL00951)

Text:

Caelestis mul-
sa ((interpunct)) propria
placabitur ara
hac divis aliis
tura merumq(ue)
damus.

Translation:

Caelestis will be appeased with her own mead. On this
altar we give to the other gods incense and wine.

Verse type:

elegiac couplet (1 hexameter + 1 pentameter)



Metadata

Identification

Reference as:	MPL00951
Title	Votive altar for Caelestis
Trismegistos-No.	179961

Editing History

Created	2026-06-29, by Eric Andreas Kunz
Work Status	Pending
Import Notice	Imported from Epigraphische Datenbank Heidelberg at 2024-07-11T12:13:24, ID there: HD043765

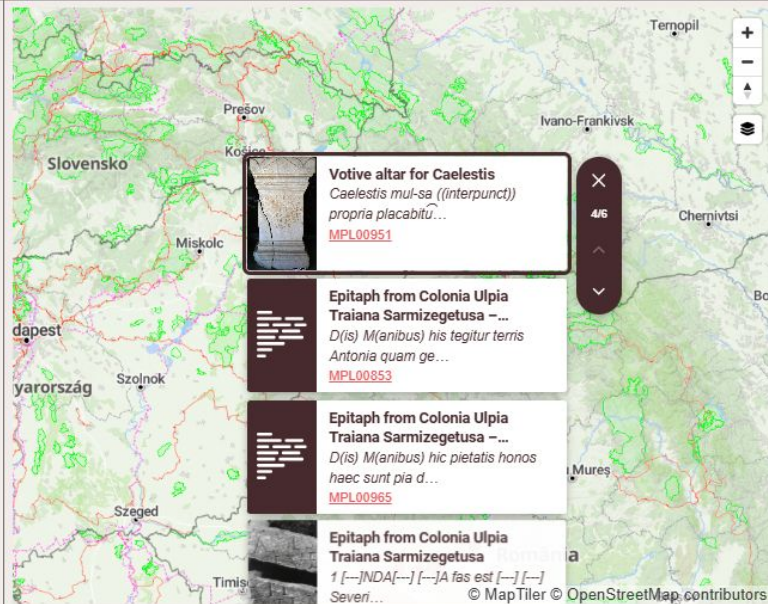
```
<titleStmt>
  <title>Votive altar for Caelestis</title>
  <respStmt>
    <resp>created by</resp>
    <name>Eric Andreas Kunz</name>
    <note>Imported from Epigraphische Datenbank Heidelberg at 2024-07-11T12:13:24, ID there: HD043765</note>
  </respStmt>
</titleStmt>
<publicationStmt>
  <authority>MAPPOLA Database</authority>
  <availability>
    <licence target="http://creativecommons.org/licenses/by-sa/4.0/">This file is licensed under the Creative Commons Attribution-ShareAlike 4.0 license.
  </licence>
</availability>
  <idno type="localID">MPL00951</idno>
  <idno type="TM">179961</idno>
  <idno type="URI">https://db.mappola.eu/r/Inscription/show/951</idno>
</publicationStmt>
```

Provenance, modern location and dating

Provenance and Location

Find Place	Colonia Ulpia Traiana Sarmizegetusa
Find Context	Found within an area sacra
Latitude	0.0
Longitude	0.0
Current Location	Sarmizegetusa, Muzeul de Arheologie
Inventory number	1341A
Current Location Details	Lapidarium

Vicinity



Places: Sarmizegetusa

[Update](#) [Cancel](#)

General

Title

Ancient Classification

Ancient Name

Province [edit list](#)

Modern Classification

Modern Name

Modern State [edit list](#)

Pleiades-ID

GeoNames-ID

Trismegistos Places-No.

Location

Longitude and Latitude coordinates for this Place. If unknown/empty use zero for both.

Latitude

Longitude

[LOAD from Pleiades](#)

Provenance, modern location and dating

Provenance and Location

Find Place	Colonia Ulpia Traiana Sarmizegetusa
Find Context	Found within an area sacra
Latitude	0.0
Longitude	0.0
Current Location	Sarmizegetusa, Muzeul de Arheologie
Inventory number	1341A
Current Location Details	Lapidarium



Places: Sarmizegetusa

Update Cancel

General

Title Sarmizegetusa

Ancient Classification

Ancient Name Sarmizegetusa

```
<history>
  <origin>
    <origPlace>
      <placeName type="ancient" ref="https://pleiades.stoa.org/places/524360868">Colonia Ulpia Traiana
Sarmizegetusa</placeName>
    </origPlace>
    <origDate notBefore-custom="231" notAfter-custom="250"
datingMethod="http://en.wikipedia.org/wiki/Julian_calendar">231-250</origDate>
  </origin>
  <provenance type="found">
    <placeName type="ancient">Colonia Ulpia Traiana Sarmizegetusa</placeName>
    <placeName type="findspot">Found within an area sacra</placeName>
    <geo>45.5147 22.7853</geo>
  </provenance>
  <provenance type="observed">Sarmizegetusa, Muzeul de Arheologie, Lapidarium</provenance>
  <provenance type="observed" subtype="squeeze-taken">no</provenance>
</history>
```

Object description

LODs from EAGLE Vocabularies
(<https://www.eagle-network.eu/resources/vocabularies/>)

Object Details	
Object Type edit list	altar
Material edit list	marble
State of Preservation edit list	complete
Dimensions	width (in cm) 36,0 height (in cm) 74,0 depth (in cm) 20,0
Letter Size in cm (min)	0,0
Letter Size in cm (max)	0,0
Execution Technique edit list	chiselled
Religion edit list	Pagan

NEW!

Fair Epigraphic Vocabularies:
<https://ontology.inscriptiones.org/>

Object description

LODs from EAGLE Vocabularies
(<https://www.eagle-network.eu/resources/vocabularies/>)

Object Details

Object Type edit list	altar
Material edit list	marble
State of Preservation edit list	complete
Dimensions	
width (in cm)	36,0
height (in cm)	74,0
depth (in cm)	20,0
Letter Size in cm (min)	0,0
Letter Size in cm (max)	0,0
Execution Technique edit list	chiselled
Religion edit list	Pagan

```
<objectDesc>
  <supportDesc>
    <support>
      <objectType ref="https://www.eagle-network.eu/voc/objtyp/lod/29">altar</objectType>
      <material ref="https://www.eagle-network.eu/voc/material/lod/48">marble</material>
      <rs type="statPreserv">complete</rs>
      <dimensions unit="cm">
        <width>36.0</width>
        <height>74.0</height>
        <depth>20.0</depth>
      </dimensions>
      <rs type="decoration" ref="none"></rs>
    </support>
  </supportDesc>
  <layoutDesc>
    <layout>
      <rs type="execution" ref="https://www.eagle-network.eu/voc/writing/lod/1">chiselled</rs>
    </layout>
  </layoutDesc>
</objectDesc>
```

Verse Layout

Prose Verse Presence

[edit list](#)

Carmen

Prose Verse Distinction

[edit list](#)

No distinction

Yes, by bigger lettering in prose

Yes, by change in lettering type

Layout Types (Prose)

[edit list](#)

Carmen figuratum

Centered

Columns

Layout Types (Verse)

[edit list](#)

Columns

Justified

Left aligned

Right aligned

Scriptio Continua in verse part

☐ N

Abbreviations in verse part

☐ N

Verse Line Correspondence

[edit list](#)

No, but follows metrical subdivisions

Graphic Signs and Spaces for Easier Carmen Reading

[edit list](#)

To mark a change in the metrical scheme - Dash

To mark a change in the metrical scheme - Other Signs

To mark a pentameter in an elegiac couplet (indentation)

Text

```
<keywords>
  <term ref="https://www.eagle-network.eu/voc/typeins/lod/79">dedication / votive inscription</term>
</keywords>
```

Text

Function edit list	dedication / votive inscription
Language(s) edit list	Italian Latin Latin written in Greek script Multilingual
Verse Type(s) edit list	Dactylic rhythm Elegiac couplet Hemiepes Other
Rhythmisation edit list	Quantitative scheme

#	Title	Parent Type
1	Saturnian	Saturnian
2	Iambo-trochaic rhythm	Iambo-trochaic rhythm
3	Lyric rhythm	Lyric rhythm
4	Unclear rhythm	Unclear rhythm
5	Dactylic rhythm	Dactylic rhythm
6	Dactylic hexameter	Dactylic rhythm
7	Elegiac couplet	Dactylic rhythm
8	Dactylic pentameter	Dactylic rhythm
9	Other	Dactylic rhythm
10	Iambic senarius	Iambo-trochaic rhythm
11	Trochaic senarius	Iambo-trochaic rhythm
12	Iambic quaternarius	Iambo-trochaic rhythm
13	Scazon	Iambo-trochaic rhythm
14	Mixed	Iambo-trochaic rhythm
15	Other	Iambo-trochaic rhythm
16	Phalaecian hendecasyllabus	Lyric rhythm
17	Other	Lyric rhythm
18	Iambic trimeter	Iambo-trochaic rhythm
19	Iambic dimeter	Iambo-trochaic rhythm
20	Trochaic septenarius	Iambo-trochaic rhythm
21	Trochaic quaternarius	Iambo-trochaic rhythm
22	Hemiepes	Dactylic rhythm
23	Glyconic	Lyric rhythm

How to encode metrical features

Line and verse

<lg> line group

</> line

Type of verse

<lg met="elegiac.couplet">

<l met="dactylic.hexameter">

<lb n="1"/>Caelestis mul

<lb n="2" break="no"/>sa propria

<lb n="3"/>placabitur ara

</l>

<l met="dactylic.pentameter">

<lb n="4"/>hac divis aliis

<lb n="5"/>tura merumq(ue)

<lb n="6"/>damus.

</l>

</lg>

EpiDoc Guidelines

Verse lines:

<https://epidoc.stoa.org/gl/latest/trans-verseline.html>

Metrical elements:

<https://epidoc.stoa.org/gl/latest/trans-metricalelements.html>

Metrical typology:

<https://epidoc.stoa.org/gl/latest/trans-metricaltypology.html>

Where to find the type of verse

MQDQ

Metrical scansion

Close

carmina epigrammata, supplementa

Permalink: <https://www.mqdq.it/texts/CE|appe|0115> [Copy](#)

AE 1993, 01345 = ILD 00282

Caelestis mulsa propria placabitur ara
Hac diuis aliis tura merumque damus

Provincia: Dacia / Località: Sarmizegetusa
Datazione: 240-260 d.C.

EDCS
EDH

Reference basis text: AE 1993, 01345
Editing of the digital edition: E. Saltelli 2010 - A. Prontera, 2022

Elegiac couplets

MusisQue DeoQue:
<https://www.mqdq.it/>

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[Export](#)

Cāelēstīs | mūlsā | prōpriā | plācābītūr āra
Hāc dīuīs ālīs || tūrā mērūmq̄e dāmūs

Text transcription with the Patrimonium converter

Enter Leiden-style text below

```
1 Caelestis·mul=H  
2 sa·propria·H  
3 placabitur·ara·H  
4 hac·divis·aliis·H  
5 tura·merumq(ue)·H  
6 damus·.¶
```

EpiDoc conversion

Epigraphic annotations

(1)▼ a á Æ A âb [a] [a?] {a} {a¹} <a> [[a]] a(bc) a(b)c(d)▼ a(bc?) a(- -) (abc) Lacuna▼ Addition▼ Illegible▼ Vacat▼ (- -) IX▼ Symbols▼ (!)

```
1 <lb n="1"/>Caelestis mul·  
2 <lb n="2" break="no"/>sa propria·  
3 <lb n="3"/>placabitur·ara·  
4 <lb n="4"/>hac divis aliis·  
5 <lb n="5"/>tura·<expan><abbr>merumq</abbr><ex>ue</ex></expan>·  
6 <lb n="6"/>damus·.¶
```

Text transcription with the Patrimonium converter

Inscription & Translations

EpiDoc

1 (lb) in word (lg) (l) a á ã â [a] [a?] {a} 'a' (!) <a> [[a]] a(b)c(d)

a(bc) a(bc?) a(---) {abc} Lacuna Addition Illegible Vacat Symbol

```
<ab>
<lg met="elegiac.couplet">
<l met="dactylic.hexameter"><lb n="1"/>Caelestis mul
<lb n="2" break="no"/>sa <seg ana="metrical"><g type="interpunct"/></seg> propria
<lb n="3"/>placabit<hi rend="ligature">ur</hi> ara</l>
<l met="dactylic.pentameter"><lb n="4"/>hac divis ali<hi rend="tall">i</hi>s
<lb n="5"/>tura <expan><abbr>merumq</abbr><ex>ue</ex></expan>
<lb n="6"/>damus.</l>
</lg>
</ab>
```

Everything looks fine!

Caelestis mul-
sa ((interpunct)) propria
placabitur ara
hac divis aliis
tura merumq(ue)
damus.

(built with the EpiDoc converter of the [Patrimonium editor](#))

convert (will override!)

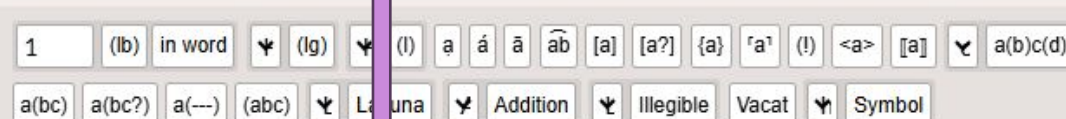
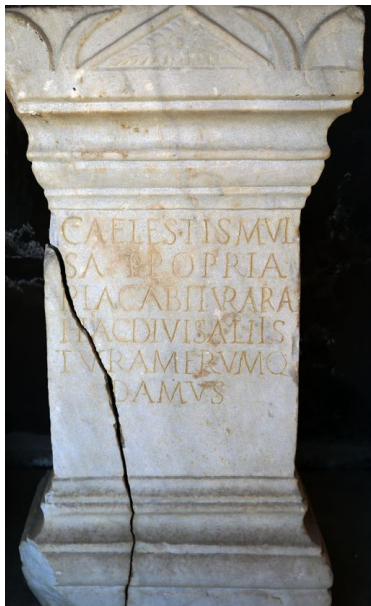
Encoding metrical features

Symbols and spaces with a semantic value

`<seg ana="metrical"><g type="interpunct"/></seg>`

Extrametrical segments

`<seg ana="extrametrical">XXX</seg>`



```
<ab>
<lg met="elegiac.couplet">
<l met="dactylic.hexameter"><lb n="1"/>Caelestis mul
<lb n="2" break="no"/>sa <seg ana="metrical"><g type="interpunct"/></seg> propria
<lb n="3"/>placabit<hi rend="ligature">ur</hi> ara</l>
<l met="dactylic.pentameter"><lb n="4"/>hac divis ali<hi rend="tall">i</hi>s
<lb n="5"/>tura <expan><abbr>merumq</abbr><ex>ue</ex></expan>
<lb n="6"/>damus.</l>
</lg>
</ab>
```

Everything looks fine!

Prosopography



MPL02248

D(is) M(anibus)

M(arci) Romani Iovini,
rhetoris eloquii Latini.

**Conditus hac Romanus
est tellure Iovinus,
docta loqui doctus
quique loqui docuit.
Manibus infernis,
si vita est gloria vitae,
vivit et hic nobis
ut Cato vel Cicero.
ut Cato vel Cicero.**

M(arcus) Iunius Severus et
Romania Marcia,
heredes, bene merenti
Fecerunt.

Prosopographical Information

Name	M. Romanus Iovinus
Gender	Male
Profession	Other profession
Role	Dedicatee

Name	M. Iunius Severus
Gender	Male
Role	Dedicator

Name	Romania Marcia
Gender	Female
Role	Dedicator

Prosopography

EpiDoc Tutorials - Names and Persons:

https://docs.google.com/presentation/d/1jzL62tv_Xmjx-H8Bdd3ejB_w_oJQJcgd8xTEAXTohQIM/present?slide=id.p

Prosopographical Information

Name	M. Romanus Iovinus
Gender	Male
Profession	Other profession
Role	Dedicatee

Name	M. Iunius Severus
Gender	Male
Role	Dedicator

Name	Romania Marcia
Gender	Female
Role	Dedicator

```
<particDesc>
  <p>
    <listPerson>
      <person gender="Male" age="none" role="Dedicatee">
        <persName>
          <name>M. Romanus Iovinus</name>
        </persName>
      </person>
      <person gender="Male" age="none" role="Dedicator">
        <persName>
          <name>M. Iunius Severus</name>
        </persName>
      </person>
      <person gender="Female" age="none" role="Dedicator">
        <persName>
          <name>Romania Marcia</name>
        </persName>
      </person>
    </listPerson>
  </p>
</particDesc>
```


A continuation, not a conclusion

- **MAPPOLA is still a work in progress:** fewer than half of the known verse inscriptions have been entered so far.
- The **next steps** include expanding the corpus, refining the encoding, improving Linked Open Data integration, and further developing the interface and design
- **Explore the database!** Search by place, metre, language, or historical context. Find an inscription that interests you and make it the starting point for your assignment.

Assignment

Exercise 2: Encode one verse inscription from the database [MAPPOLA](#).

1. Use either [Visual Studio Code](#) or [Oxygen XML](#) editor (30-day trial available) for the encoding.
2. Download and use the [EpiDoc template](#).
3. Refer to the [Text Quick Reference](#) sheet for text transcription elements, and the [Structured Data Quick Reference](#) for descriptive and historical features.
4. Make sure to encode the following features, using controlled vocabularies and geographical gazetteers when appropriate:
 - provenance
 - object type, material, and typology
 - layout features
 - in the text transcription, encode metrical features using the `<l>` and `<lg>` elements.
 - Add prosopographical markup for all people mentioned in the text
5. For the visualisation, log in to [EdEp Editor](#) using the demo account and upload your XML file.



Or try **EFES**!
(<https://github.com/EpiDoc/EFES>)

Sunoikisis Digital Classics – 30 June 2026

Between Digital Edition and Database: Digital Approaches to Signacula and Breadstamps

The SIGN-IT – Signacula in Roman and Post-Roman Italy (2nd-11th c. CE) project

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Funded by
the European Union

SIGN-IT – Signacula in Roman and Post-Roman Italy: marking religious and cultural identity (2nd-11th c. CE)

- Study of cultural and religious identity by means of marking practices
- Chronological scope: a long-term investigation from the Roman Empire to the Middle Ages.
- Geographical scope: Veneto, Emilia-Romagna, Campania, Apulia, Calabria, Sicily, Sardinia



Research Questions

- How can signacula contribute to the cultural history of Roman and Byzantine Italy?
- What role did language choice on marking tools played? Utility and/or processes of self-representation and identity construction?
- How did marking practices evolve between the Roman and Byzantine periods?

Why *signacula*?

- Signacula are stamps used to impress texts, symbols or images on different (various?) materials.
- Their functions range from authentication and ownership marking to ritual and religious use.
- Through their inscriptions and iconography, they provide access to social and cultural practices of various social strata



© Archeological Park and Museum of Altinum (Venice), inv. AL30984 (ph. Lucia Maria Orlandi)

Monogram : Θεμίστου (?) of Themistos (?).
 Saraçhane, Istanbul, end 7th c. (Kara 2017, p. 282, fig. 1)

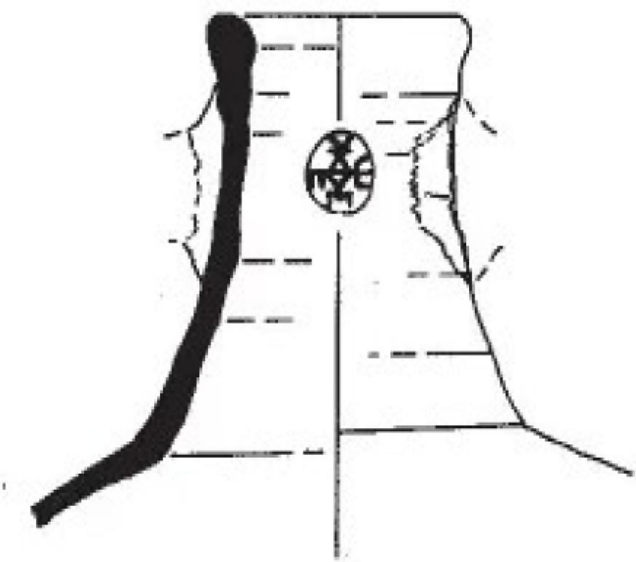


Abb. 9: Ioannes
 (aus EVANS – WIXOM 1997)

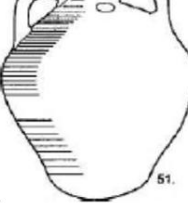
Grünbart, Lochner-
 Metaxas

« + Επὶ Πτολεμέου ἐπάρχου ».
 Under the eparch Ptolemeos
 Rhodes, b.-mid 7th c.
 (Παπανικολάου 2014, p. 172, fig. 1β)



<https://www.ancientresource.com>, n. AR3365
 (last accessed 25/06/2026)

ТИП КЛЕЙМА	ТИП АМФОРЫ	МЕСТО НАХОДКИ. ПУБЛИКАЦИЯ (место и размеры клейма в см)	АТРИБУЦИЯ
 1.	1.-тип I	1.Херсонес. Виз. Херсон. 1991, рис.6 (ручка; 2,2x1,7)	Михаил III (842-867)
 2.  3.  4.	2.-тип I; 3.-тип I(?); 4.-тип II	2.Партенит. Паршина, 1991,стр.76 , рис.6 (ручка; 2,5x1,7) 3.Херсонес. Якобсон, 1959, рис.44,4 4.Херсонес. Якобсон, 1953, рис.75,6 Визант. Херсон, 1991, №148 (ручка; 1,3x1,3)	Василий I (867-886)
 5.	5.-тип I	5.Багла. Талис, 1966, рис.1-2 (ручка; 1,4x1,5)	Евдокия, жена Василия I (889-879)
 6.  7.	6.-тип II; 7.-тип II	6.Херсонес. Якобсон 1979, рис.44,1 Якобсон 1950, рис.106 (ручка; 1,6x1,3) 7.Херсонес. Фонды ХГИАЗ №37156154 (ручка; 2,5x2,6)	Лев VI (886-912)
 8.  9.	8.-тип I;9.-тип II(?)	8.Ганос. Gunsenin, 1994,R.bc1 (ручка) 9.Константинополь. Naues,1975, fig. 27,21 (плечо, h=1,6)	Александр- соправит. Льва VI, единоличное правление (912-913)

 49.  50.  51.	 50. тип I  51.	49. Силистра. Чангова, 1959, обр. 7, 2 (плечо, d=3,4) 50. Киев. Зоценко (ст. в настоящем сборнике) (плечо; d =3,4) 51. Крым. Трал-Алушта-Судак со дна моря (частная коллекция) Киев. Зеленко (ст. в настоящ. сборнике) клеймо нечеткое (плечо; d=3,4)	Никифор III, (1078-1081)
--	--	--	-----------------------------

Parshyna 2001



†ΙΩΑΝΝΙΡ/
Α/ΣΠΑΘΑΡ/Ω
Κ/ΕΠΗΤΩΝΥ
ΚΗΑΚΟΝΑΜ/

Of John, imperial
protospatharios
and epi ton
oikeiakon, Amen

Bursa, Bithynia (10th – end 11th c.), Altinoluk 2020



Καλοκύρου πρωτοσπαθαρήω

(Seal) of Kalokyros, protospatharios

© BNF, Froehner 556 (7th /8th or 9th-10th c.)



+ Κ(ύρι)ε βοήθη τῷ σῷ δ(ούλῳ) Θεωδόρο +
Lord, help your servant Theodoros

Nomos, Auction 20, Lot 443 (9th-10th c. ?)

ΚΕΒΟΗ|Θ - ΤΩΣΩ|ΔΟΝΛΩ|ΝΗΚΗΤΑ|ΚΟΝΒΔ|ΚΛΗΧΗ|ΩΑ|ΗΗΝ+
Κ(ύρι)ε βοήθ(ει) τῷ σῷ δούλῳ Νηκήτα κουβουκλησήω ἀμῆν+
Lord, help your servant Nicetas, kouboukleisios. Amen.

Getty Museum, end 11th – beginning 12th c.
(<https://www.getty.edu/art/collection/object/1061B5>, Public Domain)



ΥΓΙΑ – ΖΩΗ
Health - Life

Live
Auctioneers,
Lot0035A_2023



4.

Apotropaic/medicinal stamp
ΥΓΙΑ – Θ(άνατος)
Health (restricting, blocking) Death

Dölger 1929



ΕΥΛΟΓΙΑ ΑΓΙΟΥ... ΑΜΗΝ (blessing of saint...
Amen) or
ΕΥΛΟΓΙΑ ΚΥΡΙΟΥ ΑΜΗΝ (blessing of the Lord
Amen).

Delphi, end 6th c.
Petridis 2012

Markers for eulogia bread, Aegina (6th-7th c.?)
Archaeological Museum of Aegina
Felten 1975



Corpus

- Preliminary estimates identify more than 1,100 known specimens for the regions under analysis → **currently the SIGN-IT DB counts 925 records, in progress**
- The corpus includes primarily preserved signacula, but can be enriched with reference to surviving impressions of lost markers.
- Integrating both categories + comparison with further sigillographic material allows a broader reconstruction of marking practices.

Current State of Research

- Studies on *signacula* have multiplied over the last two decades.
- So far, most publications remain locally or chronologically restricted.
- Multilingual corpora and long-term perspectives are largely absent.
 - Difficult dating has limited the historical use of *signacula*

Research Objectives

1. Establishing a Dating Protocol
1. Linguistic choices and Identity: mapping geographical and distribution of Latin and Greek *signacula*
1. Evolution of Marking Practices: form, function and contexts of use

Building a Digital Infrastructure

Relational structure (tables)

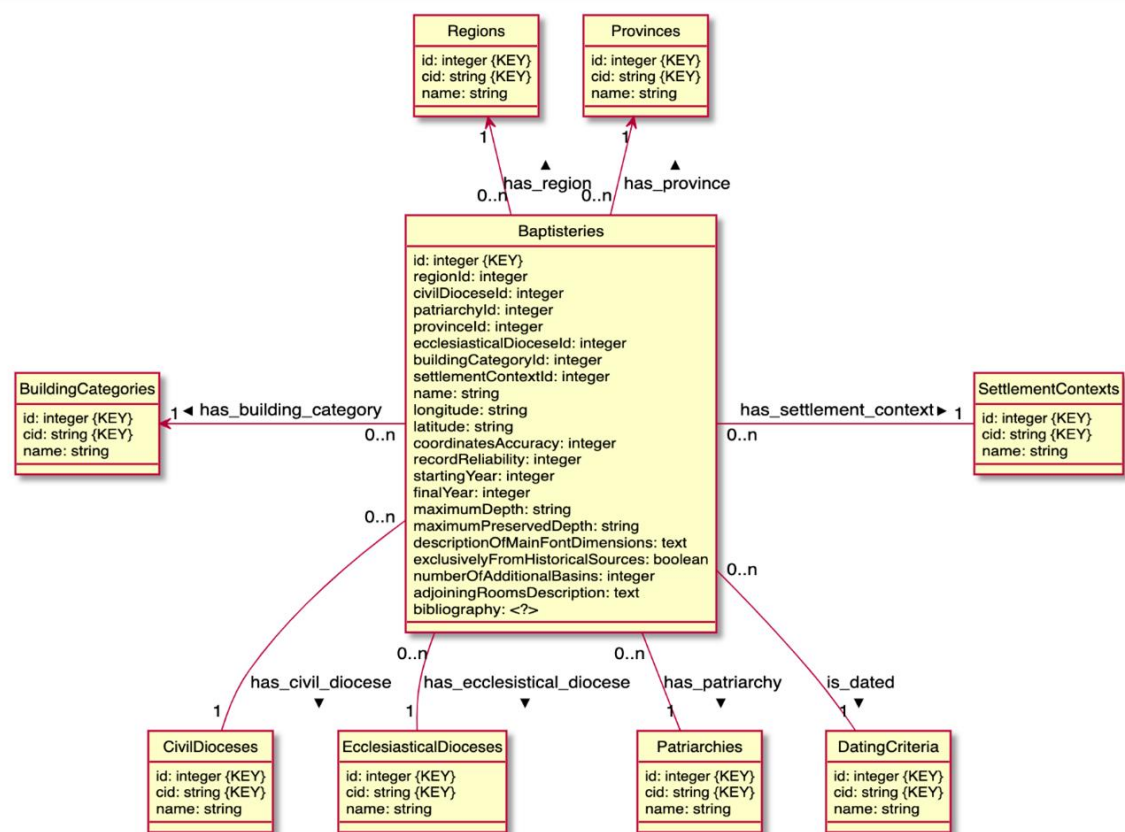


Figure 2.2 – UML de la base de données

Non-relational structure (XML)

```
<msDesc>
  <msIdentifier>
    <repository>museum/archive</repository>
    <idno>inventory number</idno>
  </msIdentifier>
  <physDesc>
    <objectDesc>
      <supportDesc>
        <support>description of object/monument (likely to include <material/>
          and <objectType/> information, <dimensions/>, etc.)</support>
      </supportDesc>
      <layoutDesc>
        <layout>description of text field/campus</layout>
      </layoutDesc>
    </objectDesc>
    <handDesc>
      <handNote>description of letters, possibly including <height>letter-heights</height>
    </handNote>
    </handDesc>
  </physDesc>
  <history>
    <origin>
      <origPlace>Place of origin</origPlace>
      <origDate>Date of origin</origDate>
    </origin>
    <provenance type="found"> Findspot and circumstances/context
    </provenance>
    <provenance type="observed"> modern location(s) (if different from repository, above)
    </provenance>
  </history>
</msDesc>
```

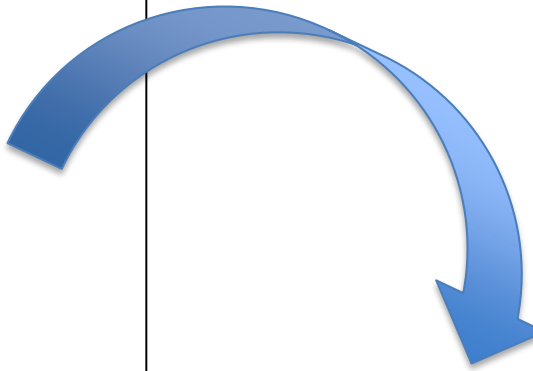
S. Stoyanova, 2019

Methodological choices underlying the model

Epigraphic information: texts, languages, abbreviations and formulae.

Morphological information: material, shape, dimensions and production techniques.

Contextual information: provenance, dating and collection history.



“Parcellation” of the information within the possibilities offered by the EpiDoc schema

Particular attention to morphology, decoration and graphic features.

Balance between analytical precision and time resources, sustainability (e.g., DC mapping)

Building a Digital Infrastructure

Why TEI/EpiDoc + SigiDoc?

- established standards for digital epigraphy and sigillography.
- Facilitates future integration with other corpora of inscribed objects.

```
● SIGN-IT_template_wComments.xml X ● SIGIT0000845_AulusTitusCaiusSaturni.xml X
TEI text
1 <?xml version="1.0" encoding="UTF-8"?>
2 <?xml-model href="tei-epidoc.rng" schematypens="http://relaxng.org/ns/structure/1.0"?>
3 <?xml-model href="tei-epidoc.rng" schematypens="http://purl.oclc.org/dsdl/schematron"?>
4 <TEI xmlns="http://www.tei-c.org/ns/1.0" xml:space="preserve" xml:lang="en">
5 <teiHeader>
6 <fileDesc>
7 <titleStmt> [20 lines]
28 <publicationStmt> [16 lines]
45 <sourceDesc>
46 <msDesc> [120 lines]
167 </sourceDesc>
168 </fileDesc>
169 <encodingDesc> [16 lines]
186 <profileDesc> [22 lines]
209 <revisionDesc><change when="2025" who="LMO">created</change> [1 line]
211 </teiHeader>
212 <facsimile> [24 lines]
237 <text>
238 <body> [40 lines]
279 </text>
280 </TEI>
```

Morphology

▼<supportDesc>

▼<support>

```
<objectType xml:lang="la" ana="signaculum" ref="https://www.eagle-network.eu/voc/objtyp/lod/240.html">Signaculum</objectType>
<material ana="bronze, copper alloy" ref="https://www.eagle-network.eu/voc/material/lod/644.html">Bronze</material>
<material type="colour" cert="unknown"/>
<rs xml:id="lamina" ana="lamina" type="shape" key="#Baratta 2014, D2a">Crescent moon with small auricle</rs>
<rs xml:id="handle" ana="handle" type="shape" key="#rectangular">Rectangular handle with circular inner ring.</rs>
<rs xml:id="bezel" ana="bezel" type="shape" key="#flattened" cert="low"/>
```

▼<p>

Total height of the object:

▼<dimensions type="object">

```
<height unit="cm"/>
```

```
<dim type="weight" unit="g"/>
```

```
</dimensions>
```

</p>

▼<p>

Dimensions of the lamina:

▼<dimensions type="lamina" subtype="profile_external">

```
<height unit="cm"/>
```

```
<width unit="cm"/>
```

```
<dim type="diameter" subtype="crescent" unit="cm">6.3</dim>
```

```
<dim type="diameter" subtype="auricle" unit="cm">3</dim>
```

```
<depth unit="cm"/>
```

```
</dimensions>
```

Morphology

```
Dimensions of the handle:
▼<dimensions type="handle">
  <height unit="cm">1.3</height>
  <width unit="cm">2.7</width>
  <depth unit="cm"/>
</dimensions>
</p>
▼<p>
  Dimensions of the bezel:
  ▼<dimensions type="bezel">
    <height unit="cm"/>
    <width unit="cm">2.7</width>
    <depth unit="cm"/>
  </dimensions>
</p>
▼<p>
  State of preservation and level of wear-and-tear:
  <rs ana="inscription" type="statPreserv" ref="https://www.eagle-network.eu/voc/statepreserv/lod/13">tit. integer</rs>
  <rs ana="relief" type="statPreserv" subtype="#wear_tear" key="#n.id."/>
  <rs ana="relief" type="original-profile" subtype="#flattened" cert="low"/>
  <rs ana="object" type="statPreserv" subtype="#restoration" key="#n.id."/>
</p>
<rs ana="authenticity" cert="high"/>
<rs ana="use" type="residue" subtype="matter" key="#concretion" rend="#medium"/>
▼<rs ana="use" type="residue" subtype="position">
  <lb n="1"/>
  Above M
  <lb n="2"/>
  above N
</rs>
</support>
</supportDesc>
```

Epigraphic features

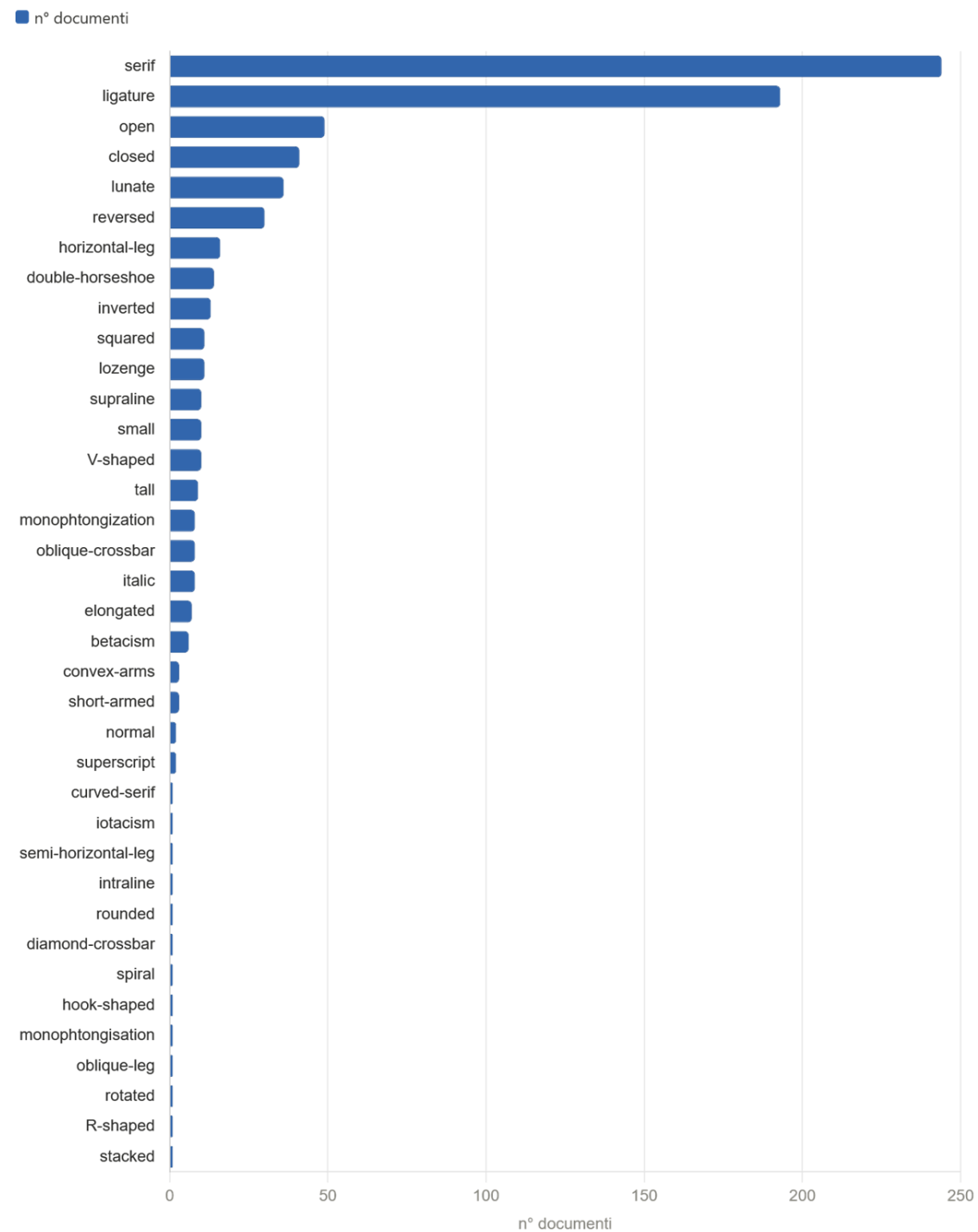
```
▼<handDesc>
  ▼<handNote ana="lamina">
    <p>Letters in relief with small horizontal serifs; litt. emin.</p>
    <lb n="1"/>
    <width unit="cm"/>
    <height unit="cm">0.8</height>
    <height ana="interlinear" unit="cm"/>
    <dim type="relief" unit="cm"/>
    <hi rend="serif">M</hi>
    <hi rend="serif">E</hi>
    <hi rend="serif">R</hi>
    <hi rend="serif">A</hi>
    <hi rend="serif">T</hi>
    <hi rend="serif">I</hi>
    <lb n="2"/>
    <width unit="cm"/>
    <height unit="cm">1.1</height>
    <height ana="interlinear" unit="cm"/>
    <dim type="relief" unit="cm"/>
    <hi rend="serif">A</hi>
    <hi rend="serif">V</hi>
    <hi rend="serif">G</hi>
    <hi rend="serif supraline">N</hi>
  </handNote>
  <handNote ana="bezel"/>
  <handNote ana="titulus-fabrilis"/>
```

Epigraphic features

```
▼<body>
  ▼<div type="edition" subtype="diplomatic">
    ▼<div type="textpart" subtype="lamina" rend="linear" style="text-direction:r-to-l">
      ▼<ab>
        <lb n="1"/>
        ▼<orig>
          <g ref="#elongated-leaf">(elongated-leaf)</g>
          MODERATI
          <g ref="#elongated-leaf">(elongated-leaf)</g>
          <lb n="2"/>
          <g ref="#ivy-leaf">(ivy-leaf)</g>
          AVG
          <hi rend="supraline">N</hi>
          <g ref="#ivy-leaf">(ivy-leaf)</g>
        </orig>
      </ab>
    </div>
  </div>
```


Initial Data Extracts

Groupings : e.g., Palaeography and Script





© Museo archeologico nazionale di Ravenna
SIGIT0000374_Caesonijs (ph. L. Orlandi)



© Museo archeologico nazionale di Ravenna
SIGIT0000362_EXOFVGNRPRS (ph. L. Orlandi)



© Museo archeologico nazionale di Napoli
SIGIT0000494_Marcus (ph. U. Soldovieri)



© Bibliothèque nationale de France
Froehner.482 (ph. Azentis)



© Bibliothèque nationale de France
SIGIT0000860_Magnos (ph. Azentis)

Context and provenance

```
</origDate>
▼<origPlace cert="high" corresp="#findspot">
  Marsicovetere - Barricella (PZ), Roman villa
  <geo decls="#WGS">40.3768327099 15.799474011</geo>
</origPlace>
</origin>
▼<provenance type="found" xml:id="findspot">
  ▼<placeName type="ancientFindspot" ref="https://pleiades.stoa.org/places/155158534" cert="high">
    Marsicovetere - Barricella (PZ), Roman villa of the Bruttii Praesentes
    <geo decls="#WGS">40.3768327099 15.799474011</geo>
  </placeName>
  ▼<placeName type="modernFindspot" ref="https://www.geonames.org/13118811/roman-villa-of-the-bruttii-praesentes.html" cert="high">
    Marsicovetere - Barricella (PZ), Roman villa of the Bruttii Praesentes
    <geo decls="#WGS">40.37667 15.79964</geo>
  </placeName>
  ▼<placeName type="locus" subtype="production" cert="high">
    Layer (US 30) located in the pars rustica of the villa, in room 8, and generated following the second earthquake that destroyed the building
    between the end of the 2nd and the beginning of the 3rd century.
    <geo decls="#WGS"> 40.376613 15.799979</geo>
  </placeName>
  <placeName type="romanProvince" ref="https://pleiades.stoa.org/places/991362" cert="high">Regio III - Lucania et Bruttii</placeName>
  ▼<placeName type="modernRegion" ref="https://www.geonames.org/3182306/basilicata.html" cert="high">
    Basilicata
    <geo decls="#WGS">40.5041 16.11396</geo>
  </placeName>
  ▼<placeName type="modernCountry" ref="https://www.geonames.org/3175395/italy.html" cert="high">
    Italy
    <geo decls="#WGS">42.83333 12.83333</geo>
  </placeName>
</provenance>
```

```
▼<div type="edition" subtype="editorial" xml:lang="la">
  ▼<div type="textpart" subtype="lamina">
    ▼<ab>
      <lb n="1"/>
      ▼<persName type="attested">
        ▼<name type="simplex-nomen" nymRef="#sn.Moderatus" xml:id="SIGIT0000001_text_1">
          <w ana="genitive">Moderati</w>
        </name>
      </persName>
      <lb n="2"/>
      ▼<s type="formula" function="possessive" subtype="aug-n" xml:id="SIGIT0000001_text_2">
        ▼<rs type="officeCivil" ref="Augustus" xml:id="SIGIT0000001_text_3">
          ▼<expan>
            <abbr>Aug</abbr>
            <ex>usti</ex>
          </expan>
        </rs>
        ▼<w ana="genitive">
          ▼<expan>
            <abbr>n</abbr>
            <ex>ostri</ex>
          </expan>
        </w>
        ▼<w ana="genitive">
          <supplied reason="subaudible" sameAs="#SIGIT0000001_text_1">servi</supplied>
        </w>
      </s>
    </ab>
  </div>
</div>
```



```

<textClass>
  <keywords>
    <term ref="http://www.eagle-network.eu/voc/typeins/lod/106">cetera</term>
    <term key="#textType_3" cert="low">holder_delegated_slave</term>
  </keywords>
</textClass>
<particDesc>
  <person gender="masculine" role="libertus" cert="high"><lb n="1" corresp="#lamina"/><persName>Domitius Cytisus (?)</persName>
  <idno type="PIR" corresp="URI"/>
  <idno type="PBW" corresp="URI"/>
</person>
  <person gender="masculine" role="slave" cert="high"><lb n="1" corresp="#bezel"/><persName>Calvinus</persName>
  <idno type="PIR" corresp="URI"/>
  <idno type="PBW" corresp="URI"/>
</person>
</particDesc>

```

SIGN-IT

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Search



Items

1 of 1



1-25 of 25

[Advanced search](#)

Created ▾

Descending ▾

Sort

Signaculum of Lucius Antonius Hermes



Signaculum (of) Amicus



Signaculum of Aemilius Alexander



Signaculum of Lucius Antonius Hermes



Title
Signaculum of Lucius Antonius Hermes

Contributor
Lucia Maria Orlandi

Publisher
European Union (MSCA-PF-EF n. 101146566)

Identifier
SIGIT0000366

Rights

Title
Signaculum of Lucius Antonius Hermes

Contributor
Lucia Maria Orlandi

Publisher
European Union (MSCA-PF-EF n. 101146566)

Identifier
SIGIT0000366

Rights
Licensed under a Creative Commons-Attribution 4.0 licence.

Rights Holder
Museo archeologico nazionale

Language
Latin

Type
Signaculum

Format
Total height of the object: height - 2.4 cm; weight - 53.36 g;
Dimensions of the lamina: height - 2.4 cm; width - 6.7 cm; depth - 0.4 cm;
Dimensions of the border: width - 0.2cm;
Dimensions of the handle: height - 1.8 cm; width - 2.7 cm; depth - 0.4 cm;
Dimensions of the bezel: height - 0.7 cm; width - 1.9 cm; depth - 0.2 cm;

Date Created
1 C.E - 200 C.E.

Subject
Lucius Antonius Hermes

Has Part
Diplomatic edition:
L(ivy-leaf)ANTONI
HERMETIS(ivy-leaf)
Editorial edition:
Luci Antoni Hermetis Luci Antoni Hermetis
Translation:
(Signaculum) of Lucius Antonius Hermes

Bibliographic Citation
Edition:
CIL XI 6712, 44
Discussion:
(2) Rigato, Daniela. «I signacula ex aere del Museo Nazionale di Ravenna: un quadro introduttivo.» Instrumenta inscripta V. Signacula ex aere. Aspetti epigrafici, archeologici, giuridici, prosopografici, collezionistici. Atti del Convegno (Verona, 2012), Scienze e Lettere, 2014, pp. 203-216, p. 207, fig.2

References
Export-XML

Conclusions & Perspectives

- First cross-period, multilingual corpus of signacula from Roman and Byzantine Italy (and beyond...)
- Based on existing standard (EpiDoc, SigiDoc), with enhanced granularity to suit the specific nature of “inscribed objects”
- Transferable model
- Common and accessible platform for the study of marking practices
- Stay tuned...

Assignment

Where should you start when developing a cataloguing scheme?

Keep the following in mind:

- what you want to know from the objects,
- BUT ALSO
- that new questions may arise in the future.

Have a look at Figures 1-8 of the scientific blog post by Dr Ilenia Gradante, [here](#).

- identify as many features that describe stamps (shape, dimensions, text...)
- organise them by categories